Lab 5

Carson Large 2477823
Cl808@uw.edu

1. In the notebook, the audio waveform is transformed into time-frequency features before being passed to the keyword spotting model. What is the purpose of converting raw audio into spectrogram/MFCC-style features instead of directly using the raw time-domain waveform?

You need the audio to be expressed as an image so you can take advantage of 2D CNN

2. During full integer quantization, the notebook uses a representative dataset. What is the purpose of the representative dataset, and why is it important for creating an INT8 TensorFlow Lite model?

It needs to have a reference for calibrating the values

3. What does `audio_processor.get_data()` do? Refer to `input_data.py`.

It fetches a batch of labeled audio samples, preprocesses them into model input features, optionally applies augmentation, and returns both the feature data and labels.

4. In the final model export step, the TensorFlow Lite model is converted into a C/C++ byte array. What are `g_model`and `g_model_len`, and why do they need to be copied correctly into the Arduino source file?

G_model is the model itself expressed as c/c++, g_model_len is the exact length of the model and tensorflow on arduino needs both exactly to interpret the model.

5. Please include a screenshot showing the correct identification of at least one keyword, either **"YES"** or **"NO"**, as displayed in the Arduino IDE Serial Monitor after deploying your KWS model on the Arduino Nano 33 BLE. The screenshot should be included in your submitted PDF.

micro_speech - micro_features_model.cpp | Arduino IDE 2.3.8

File Edit Sketch Tools Help

Arduino Nano 33 BLE

es_generator.h micro_features_micro_model_settings.cpp micro_features_micro_moc

```
#endif

const unsigned char g_model[] DATA_ALIGN_ATTRIBUTE = {
```

Output Serial Monitor

Message (Enter to send message to 'Arduino Nano 33 BLE' on 'COM3')

```
Heard unknown (202) @45424ms
Heard unknown (201) @47648ms
Heard unknown (226) @49200ms
Heard unknown (202) @51136ms
Heard unknown (201) @54272ms
Heard unknown (218) @56112ms
Heard unknown (208) @57680ms
Heard unknown (202) @61184ms
Heard unknown (211) @64672ms
Heard unknown (220) @66800ms
Heard unknown (245) @68368ms
Heard yes (212) @71120ms
Heard unknown (216) @72144ms
Heard unknown (247) @73696ms
Heard unknown (202) @76560ms
Heard no (203) @78864ms
Heard unknown (202) @80240ms
Heard unknown (214) @83840ms
Heard unknown (253) @85392ms
Heard unknown (211) @87792ms
Heard unknown (239) @89360ms
Heard unknown (210) @94416ms
Heard unknown (253) @95984ms
Heard unknown (225) @97552ms
Heard unknown (249) @99120ms
Heard unknown (207) @101520ms
```